

확률적 계획법 HW 2

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Problem 2.3 Quantile Deviation Risk Measure

Prove that quantile deviation (QDEV) is not coherent by showing that it does not satisfy the properties of translation invariance and monotonicity but satisfies positive homogeneity and convexity.

$$\phi_{\text{QDEV}}^{\varepsilon_1, \varepsilon_2}(x) = \min_{\eta \in \mathbb{R}} \mathbb{E} [\varepsilon_1 (\eta - f(x, \tilde{\omega}))_+ + \varepsilon_2 (f(x, \tilde{\omega}) - \eta)_+]$$

Translation Invariance

Let $f(x, \tilde{\omega}) \in \mathcal{F}$, $a \in \mathbb{R}$, and $\eta' = \eta - a$.

$$\begin{aligned} & \min_{\eta \in \mathbb{R}} \mathbb{E} [\varepsilon_1 (\eta - (f(x, \tilde{\omega}))_+^a)_+ + \varepsilon_2 ((f(x, \tilde{\omega}))_+^a - \eta)_+] \\ &= \min_{\eta \in \mathbb{R}} \mathbb{E} [\varepsilon_1 (\eta - a - f(x, \tilde{\omega}))_+ + \varepsilon_2 (f(x, \tilde{\omega}) + a - \eta)_+] \\ &= \min_{\eta' \in \mathbb{R}} \mathbb{E} [\varepsilon_1 (\eta' - f(x, \tilde{\omega}))_+ + \varepsilon_2 (f(x, \tilde{\omega}) - \eta')_+] \end{aligned}$$

If $a \neq 0$, the last term is not equal to $a + \min_{\eta \in \mathbb{R}} \mathbb{E} [\varepsilon_1 (\eta - f(x, \tilde{\omega}))_+ + \varepsilon_2 (f(x, \tilde{\omega}) - \eta)_+]$

$\Rightarrow (\because)$ QDEV does not satisfy translation invariance. ■

Monotonicity

Let $f(x, \tilde{\omega}_1), f(x, \tilde{\omega}_2) \in \mathcal{F}$, and $f(x, \tilde{\omega}_1) \leq f(x, \tilde{\omega}_2)$.

Then, $[\eta - f(x, \tilde{\omega}_1)]_+ \geq [\eta - f(x, \tilde{\omega}_2)]_+$ and $[f(x, \tilde{\omega}_1) - \eta]_+ \leq [f(x, \tilde{\omega}_2) - \eta]_+$

Therefore,

$$\min_{\eta \in \mathbb{R}} \mathbb{E} [\varepsilon_1 (\eta - f(x, \tilde{\omega}_1))_+ + \varepsilon_2 (f(x, \tilde{\omega}_1) - \eta)_+] \leq \min_{\eta \in \mathbb{R}} \mathbb{E} [\varepsilon_1 (\eta - f(x, \tilde{\omega}_2))_+ + \varepsilon_2 (f(x, \tilde{\omega}_2) - \eta)_+]$$

may not hold for $\forall \varepsilon_1, \varepsilon_2$.

$\Rightarrow (\because)$ QDEV does not satisfy monotonicity. ■

Prob. 2.3. ok

Positive Homogeneity

Let $f(x, \tilde{\omega}) \in \mathcal{F}$, $c > 0$, and $\eta' = \frac{\eta}{c}$, then we have

$$\begin{aligned} & \min_{\eta \in \mathbb{R}} \mathbb{E} [\varepsilon_1 (\eta - c f(x, \tilde{\omega}))_+ + \varepsilon_2 (c f(x, \tilde{\omega}) - \eta)_+] \\ &= \min_{\eta \in \mathbb{R}} \mathbb{E} [c \varepsilon_1 (\frac{\eta}{c} - f(x, \tilde{\omega}))_+ + c \varepsilon_2 (f(x, \tilde{\omega}) - \frac{\eta}{c})_+] \\ &= c \min_{\eta' \in \mathbb{R}} \mathbb{E} [\varepsilon_1 (\eta' - f(x, \tilde{\omega}))_+ + \varepsilon_2 (f(x, \tilde{\omega}) - \eta')_+] \end{aligned}$$

$\Rightarrow (\circ \circ)$ QDEV satisfies positive homogeneity. ■

Convexity

Let $f(x, \tilde{\omega}_1), f(x, \tilde{\omega}_2) \in \mathcal{F}$.

Note that for $\forall \eta \in \mathbb{R}$, $(\eta - x)_+ = \max(\eta - x, 0)$ and ... ①

$(x - \eta)_+ = \max(x - \eta, 0)$ are convex functions. ... ②

Then, for $\forall \beta \in [0, 1]$,

$$(\eta - (\beta f(x, \tilde{\omega}_1) + (1-\beta)f(x, \tilde{\omega}_2)))_+ \leq \beta (\eta - f(x, \tilde{\omega}_1))_+ + (1-\beta) (\eta - f(x, \tilde{\omega}_2))_+ \quad \text{holds by ①,}$$

$$((\beta f(x, \tilde{\omega}_1) + (1-\beta)f(x, \tilde{\omega}_2)) - \eta)_+ \leq \beta (f(x, \tilde{\omega}_1) - \eta)_+ + (1-\beta) (f(x, \tilde{\omega}_2) - \eta)_+ \quad \text{holds by ②.}$$

$$\begin{aligned} & \Rightarrow \mathbb{E} [\varepsilon_1 (\eta - (\beta f(x, \tilde{\omega}_1) + (1-\beta)f(x, \tilde{\omega}_2)))_+ + \varepsilon_2 ((\beta f(x, \tilde{\omega}_1) + (1-\beta)f(x, \tilde{\omega}_2)) - \eta)_+] \\ & \leq \mathbb{E} [\varepsilon_1 (\beta (\eta - f(x, \tilde{\omega}_1))_+ + (1-\beta) (\eta - f(x, \tilde{\omega}_2))_+) + \varepsilon_2 (\beta (f(x, \tilde{\omega}_1) - \eta)_+ + (1-\beta) (f(x, \tilde{\omega}_2) - \eta)_+)] \\ & = \beta \mathbb{E} [\varepsilon_1 (\eta - f(x, \tilde{\omega}_1))_+ + \varepsilon_2 (f(x, \tilde{\omega}_1) - \eta)_+] + (1-\beta) \mathbb{E} [\varepsilon_1 (\eta - f(x, \tilde{\omega}_2))_+ + \varepsilon_2 (f(x, \tilde{\omega}_2) - \eta)_+] \end{aligned}$$

$\Rightarrow (\circ \circ)$ By taking the minimum over $\forall \eta \in \mathbb{R}$, we can find that QDEV satisfies convexity. ■

$\Rightarrow (\circ \circ)$

QDEV satisfies positive homogeneity and convexity
but does not satisfy translation invariance and monotonicity.

Therefore, QDEV is not coherent. ■

Problem 2.4 Absolute Semideviation Risk Measure

Prove that absolute semideviation (ASD) is not coherent by showing that it does not satisfy the properties of translation invariance and monotonicity but satisfies positive homogeneity and convexity.

$$\phi_{ASD}(X) = \mathbb{E} \left[\left(f(X, \tilde{\omega}) - \mathbb{E}[f(X, \tilde{\omega})] \right)_+ \right]$$

Translation Invariance

Let $f(X, \tilde{\omega}) \in \mathcal{F}$, $a \in \mathbb{R}$, then

$$\begin{aligned} \mathbb{E} \left[\left(f(X, \tilde{\omega}) + a - \mathbb{E}[f(X, \tilde{\omega}) + a] \right)_+ \right] &= \mathbb{E} \left[\left(f(X, \tilde{\omega}) + a - a - \mathbb{E}[f(X, \tilde{\omega})] \right)_+ \right] \\ &= \mathbb{E} \left[\left(f(X, \tilde{\omega}) - \mathbb{E}[f(X, \tilde{\omega})] \right)_+ \right] \end{aligned}$$

If $a \neq 0$, then the last term is not equal to $a + \mathbb{E} \left[\left(f(X, \tilde{\omega}) - \mathbb{E}[f(X, \tilde{\omega})] \right)_+ \right]$
 $\Rightarrow$ (s.) ASD does not satisfy translation invariance. ■

Monotonicity

Ex: Let $f(X, \tilde{\omega}_1), f(X, \tilde{\omega}_2) \in \mathcal{F}$

$$\text{Let } f(X, \tilde{\omega}_1) = \begin{cases} 0 & \text{with prob. } 0.5 \\ 1 & \text{with prob. } 0.5 \end{cases}$$

$$\text{Let } f(X, \tilde{\omega}_2) = \begin{cases} 1 & \text{with prob. } 0.1 \\ 2 & \text{with prob. } 0.9 \end{cases}$$

$$\Rightarrow f(X, \tilde{\omega}_1) \leq f(X, \tilde{\omega}_2)$$

$$\text{Note that } \mathbb{E}[f(X, \tilde{\omega}_1)] = 0.5 \text{ and } \mathbb{E}[f(X, \tilde{\omega}_2)] = 1.9$$

$$\Rightarrow \left(f(X, \tilde{\omega}_1) - \mathbb{E}[f(X, \tilde{\omega}_1)] \right)_+ = \begin{cases} (0 - 0.5)_+ = 0 & \text{with prob. } 0.5 \\ (1 - 0.5)_+ = 0.5 & \text{with prob. } 0.5 \end{cases}$$

$$\Rightarrow \left(f(X, \tilde{\omega}_2) - \mathbb{E}[f(X, \tilde{\omega}_2)] \right)_+ = \begin{cases} (1 - 1.9)_+ = 0 & \text{with prob. } 0.1 \\ (2 - 1.9)_+ = 0.1 & \text{with prob. } 0.9 \end{cases}$$

$$\Rightarrow \mathbb{E} \left[\left(f(X, \tilde{\omega}_1) - \mathbb{E}[f(X, \tilde{\omega}_1)] \right)_+ \right] = 0 \times 0.5 + 0.5 \times 0.5 = 0.25 \geq \mathbb{E} \left[\left(f(X, \tilde{\omega}_2) - \mathbb{E}[f(X, \tilde{\omega}_2)] \right)_+ \right] = 0 \times 0.1 + 0.1 \times 0.9 = 0.09$$

$\Rightarrow$ (s.) ASD does not satisfy Monotonicity. ■

Prob. 2.4. 01/14

Positive Homogeneity

Let $f(x, \omega) \in \mathcal{F}$, $c > 0$

$$\mathbb{E} [[cf(x, \omega) - \mathbb{E}[cf(x, \omega)]]_+] = \mathbb{E} [[cf(x, \omega) - c\mathbb{E}[f(x, \omega)]]_+]$$

$$= \mathbb{E} [c [f(x, \omega) - \mathbb{E}[f(x, \omega)]]_+] = c \mathbb{E} [[f(x, \omega) - \mathbb{E}[f(x, \omega)]]_+]$$

$\Rightarrow$ (i.) ASD satisfies positive homogeneity. ■

Convexity

Let $f(x, \omega_1), f(x, \omega_2) \in \mathcal{F}$

Note that $(x)_+ = \max(x, 0)$ is a convex function. ... ③

then, for $\forall \beta \in [0, 1]$,

$$\mathbb{E} [(\beta f(x, \omega_1) + (1-\beta) f(x, \omega_2) - \mathbb{E} [\beta f(x, \omega_1) + (1-\beta) f(x, \omega_2)])_+]$$

$$= \mathbb{E} [(\beta f(x, \omega_1) + (1-\beta) f(x, \omega_2) - \beta \mathbb{E}[f(x, \omega_1)] - (1-\beta) \mathbb{E}[f(x, \omega_2)])_+]$$

$$= \mathbb{E} [(\beta (f(x, \omega_1) - \mathbb{E}[f(x, \omega_1)]) + (1-\beta) (f(x, \omega_2) - \mathbb{E}[f(x, \omega_2)]))_+]$$

$$\leq \mathbb{E} [\beta (f(x, \omega_1) - \mathbb{E}[f(x, \omega_1)])_+ + (1-\beta) (f(x, \omega_2) - \mathbb{E}[f(x, \omega_2)])_+] \quad \text{by ③}$$

$$= \beta \mathbb{E} [(f(x, \omega_1) - \mathbb{E}[f(x, \omega_1)])_+] + (1-\beta) \mathbb{E} [(f(x, \omega_2) - \mathbb{E}[f(x, \omega_2)])_+]$$

$\Rightarrow$ (i.) ASD satisfies convexity. ■

$\Rightarrow$ (ii) ASD satisfies positive homogeneity and convexity
but does not satisfy translation invariance and monotonicity.

Therefore, ASD is not coherent. ■

Problem: DEP of MR-SP with Absolute Semideviation

Show how to derive the deterministic equivalent formulation (DEP) for the MR-SP with *Absolute Semideviation*, which is described on page 39 of the Chapter 02 lecture note. Note that you are required to show how to derive DEP from their general MR-SP formulation.

$$\min_{x \in X} \mathbb{E}[f(x, \tilde{\omega})] + \lambda \phi_{ASD}(x)$$

$$\begin{aligned} &= \min_{x \in X} \mathbb{E}[f(x, \tilde{\omega})] + \lambda \mathbb{E}[\max\{f(x, \tilde{\omega}) - \mathbb{E}[f(x, \tilde{\omega})], 0\}] \\ &= \min_{x \in X} \mathbb{E}[f(x, \tilde{\omega})] + \lambda (\mathbb{E}[\max\{f(x, \tilde{\omega}), \mathbb{E}[f(x, \tilde{\omega})]\}] - \mathbb{E}[f(x, \tilde{\omega})]) \end{aligned}$$

$$\text{Let } f(x, \omega) = c^T x + q(\omega)^T y(\omega)$$

$$\mathbb{E}[f(x, \tilde{\omega})] = c^T x + \sum_{\bar{\omega} \in \Omega} p(\bar{\omega}) q(\bar{\omega})^T y(\bar{\omega})$$

$$\mathbb{E}[\max\{f(x, \tilde{\omega}), \mathbb{E}[f(x, \tilde{\omega})]\}] = \min \sum_{\omega \in \Omega} p(\omega) v(\omega)$$

$$\text{s.t. } v(\omega) \geq f(x, \omega) \quad \forall \omega \in \Omega$$

$$v(\omega) \geq \mathbb{E}[f(x, \tilde{\omega})] \quad \forall \omega \in \Omega$$

Therefore, DEP form for the MR-SP with ASD is :

$$\min (1-\lambda) c^T x + (1-\lambda) \sum_{\omega \in \Omega} p(\omega) q(\omega)^T y(\omega) + \lambda \sum_{\omega \in \Omega} p(\omega) v(\omega)$$

$$\text{s.t. } c^T x + q(\omega)^T y(\omega) \geq v(\omega), \quad \forall \omega \in \Omega$$

$$-c^T x - q(\omega)^T y(\omega) + v(\omega) \geq 0, \quad \forall \omega \in \Omega$$

$$-c^T x - \sum_{\bar{\omega} \in \Omega} p(\bar{\omega}) q(\bar{\omega})^T y(\bar{\omega}) + v(\omega) \geq 0, \quad \forall \omega \in \Omega$$

$$x \in X, y(\omega) \in \mathbb{R}_+^{n_2}, v(\omega) \in \mathbb{R}, \quad \forall \omega \in \Omega$$

Problem: DEP of MR-SP with Central Deviation

Show how to derive the deterministic equivalent formulation (DEP) for the MR-SP with *Central Deviation*, which is described on page 40 of the Chapter 02 lecture note. Note that you are required to show how to derive DEP from their general MR-SP formulation.

$$\begin{aligned}
 & \mathbb{E}[f(x, \tilde{\omega})] + \lambda \phi_{\text{CDV}}(x) \\
 = & \mathbb{E}[f(x, \tilde{\omega})] + \lambda \mathbb{E}[|f(x, \tilde{\omega}) - \mathbb{E}[f(x, \tilde{\omega})]|] \\
 = & \mathbb{E}[f(x, \tilde{\omega})] + \lambda (\mathbb{E}[(f(x, \tilde{\omega}) - \mathbb{E}[f(x, \tilde{\omega})])_+] + \mathbb{E}[(\mathbb{E}[f(x, \tilde{\omega})] - f(x, \tilde{\omega}))_+]) \\
 = & \mathbb{E}[f(x, \tilde{\omega})] + \lambda (\mathbb{E}[\text{Max}\{f(x, \tilde{\omega}), \mathbb{E}[f(x, \tilde{\omega})]\}] - \mathbb{E}[f(x, \tilde{\omega})] \\
 & + \mathbb{E}[\text{Max}\{\mathbb{E}[f(x, \tilde{\omega})], f(x, \tilde{\omega})\} - f(x, \tilde{\omega})]) \\
 = & (1 - 2\lambda) \mathbb{E}[f(x, \tilde{\omega})] + 2\lambda \mathbb{E}[\text{Max}\{f(x, \tilde{\omega}), \mathbb{E}[f(x, \tilde{\omega})]\}]
 \end{aligned}$$

$$\begin{aligned}
 \text{Let } f(x, \omega) &= c^T x + z(\omega)^T y(\omega) \\
 \mathbb{E}[f(x, \tilde{\omega})] &= c^T x + \sum_{\tilde{\omega} \in \Omega} p(\tilde{\omega}) \bar{z}(\tilde{\omega})^T \bar{y}(\tilde{\omega}) \\
 \mathbb{E}[\text{Max}\{f(x, \tilde{\omega}), \mathbb{E}[f(x, \tilde{\omega})]\}] &= \text{Min} \sum_{\omega \in \Omega} p(\omega) v(\omega) \\
 \text{s.t. } v(\omega) &\geq f(x, \omega) \quad , \quad \forall \omega \in \Omega \\
 v(\omega) &\geq \mathbb{E}[f(x, \tilde{\omega})] \quad , \quad \forall \omega \in \Omega
 \end{aligned}$$

Therefore, DEP form for the MR-SP with Central Deviation is :

$$\begin{aligned}
 & \text{Min } (1 - 2\lambda) c^T x + (1 - 2\lambda) \sum_{\omega \in \Omega} p(\omega) \bar{z}(\omega)^T \bar{y}(\omega) + 2\lambda \sum_{\omega \in \Omega} p(\omega) v(\omega) \\
 \text{s.t. } & T(\omega)x + W y(\omega) \geq r(\omega) \quad , \quad \forall \omega \in \Omega \\
 & -c^T x - \bar{z}(\omega)^T \bar{y}(\omega) + v(\omega) \geq 0 \quad , \quad \forall \omega \in \Omega \\
 & -c^T x - \sum_{\tilde{\omega} \in \Omega} p(\tilde{\omega}) \bar{z}(\tilde{\omega})^T \bar{y}(\tilde{\omega}) + v(\omega) \geq 0 \quad , \quad \forall \omega \in \Omega
 \end{aligned}$$

$$x \in X, y(\omega) \in \mathbb{R}_+^{n_2}, v(\omega) \in \mathbb{R} \quad , \quad \forall \omega \in \Omega$$